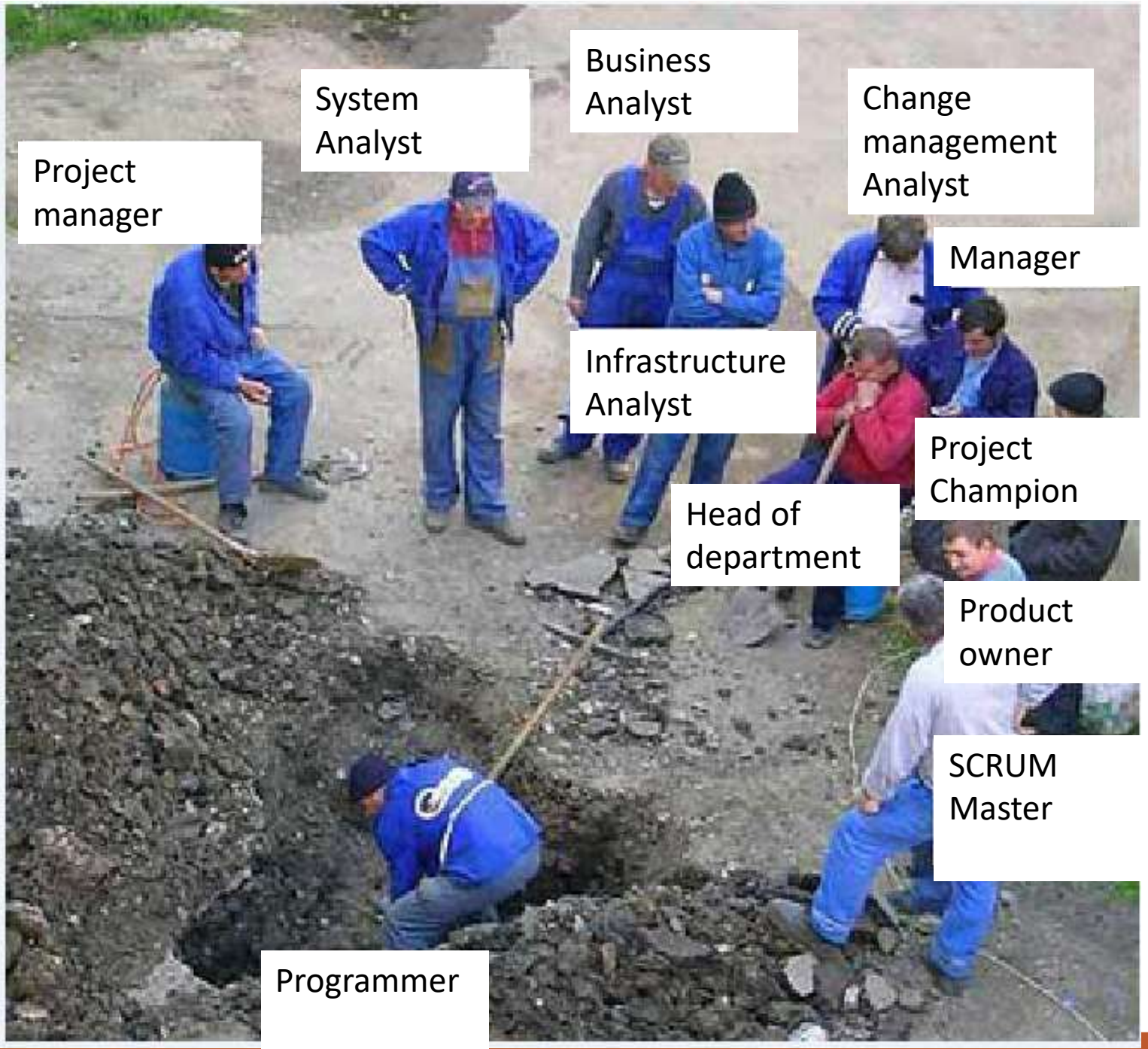
System Analysis

DR. ANDREI DRAGUNOV

How course will be organized?

- Telegram channel

  (find in eclass)
- Every week I'll upload one or two lectures
- Send me email should you have any questions
- Will be two exams: Midterm and final (plus two quizzes)
- After final exam I can give you recommendation letter only if your grade will be higher than B+ (B+ or A0 or A+)



Role	Responsibilities
Business analyst	Analyzing the key business aspects of the system
	Identifying how the system will provide business value
	Designing the new business processes and policies
Systems analyst	Identifying how technology can improve business processes
	Designing the new business processes
	Designing the information system
	Ensuring that the system conforms to information systems standards
Infrastructure analyst	Ensuring the system conforms to infrastructure standards
	Identifying infrastructure changes needed to support the system
hange management nalyst	Developing and executing a change management plan
	Developing and executing a user training plan
Project manager	Managing the team of analysts, programmers, technical writers, and other specialists
	Developing and monitoring the project plan
	Assigning resources
	Serving as the primary point of contact for the project

Introduction to Systems Analysis and Design

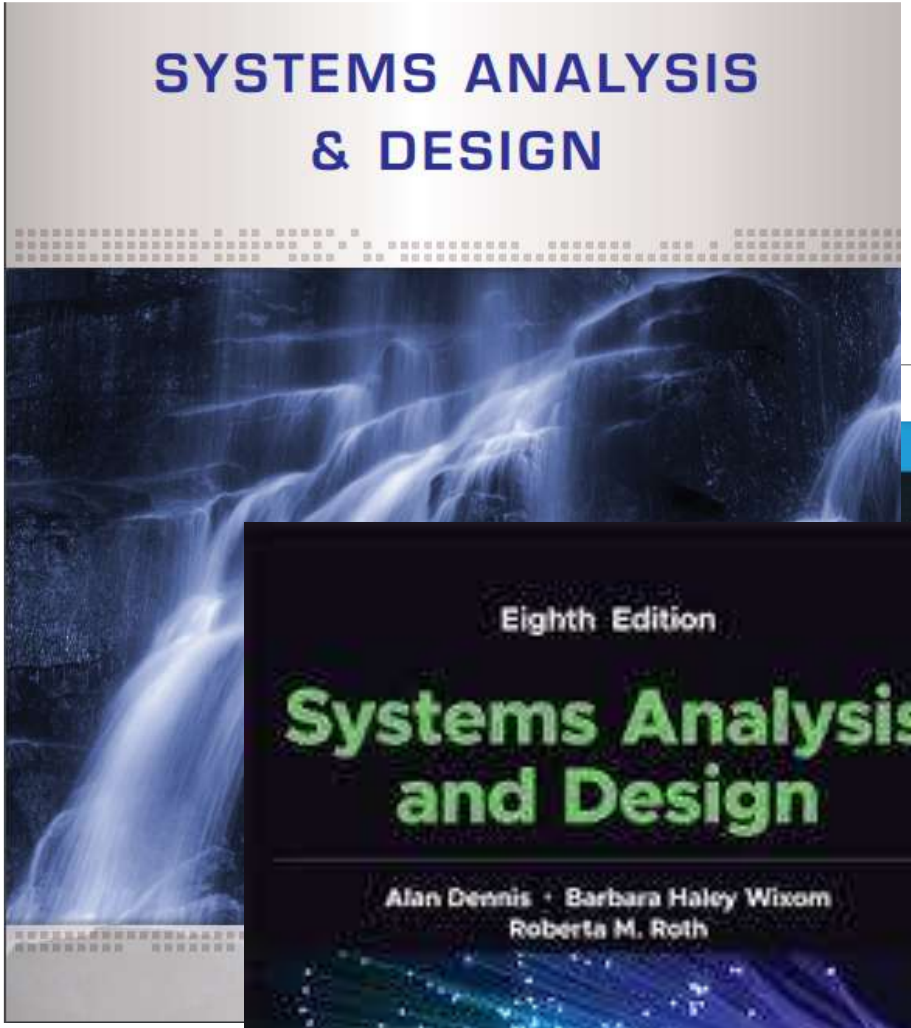
Why do we need a formal process?

- Failures occur (too) often
- Creating systems is not intuitive
- Projects are late, over budget or delivered with fewer features than planned

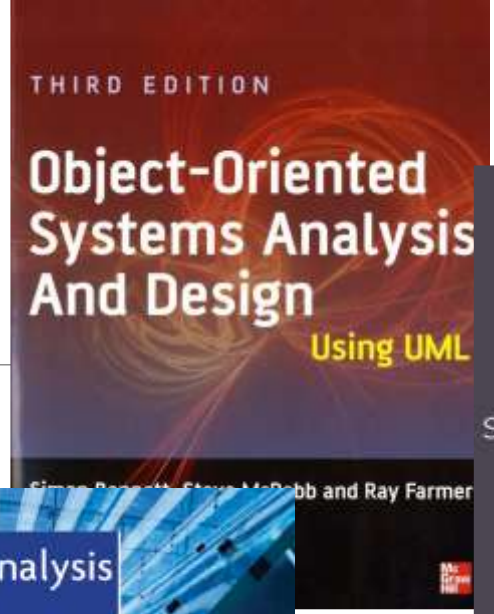
The System Analyst is the key person

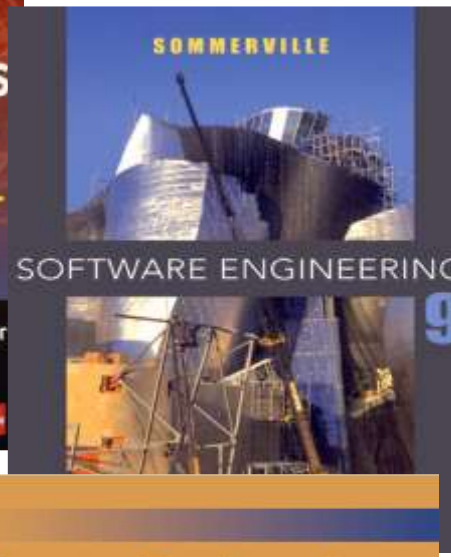
- Designs a system to add value
- Must understand the business processes
- Job is rewarding, yet challenging
- Requires specific skill sets

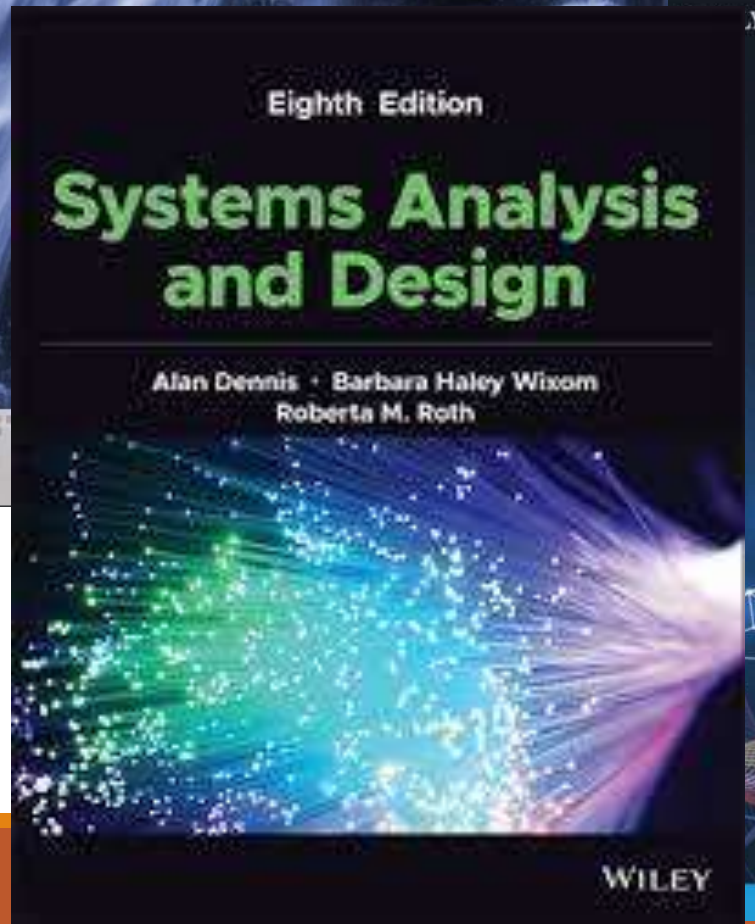
Syllabus

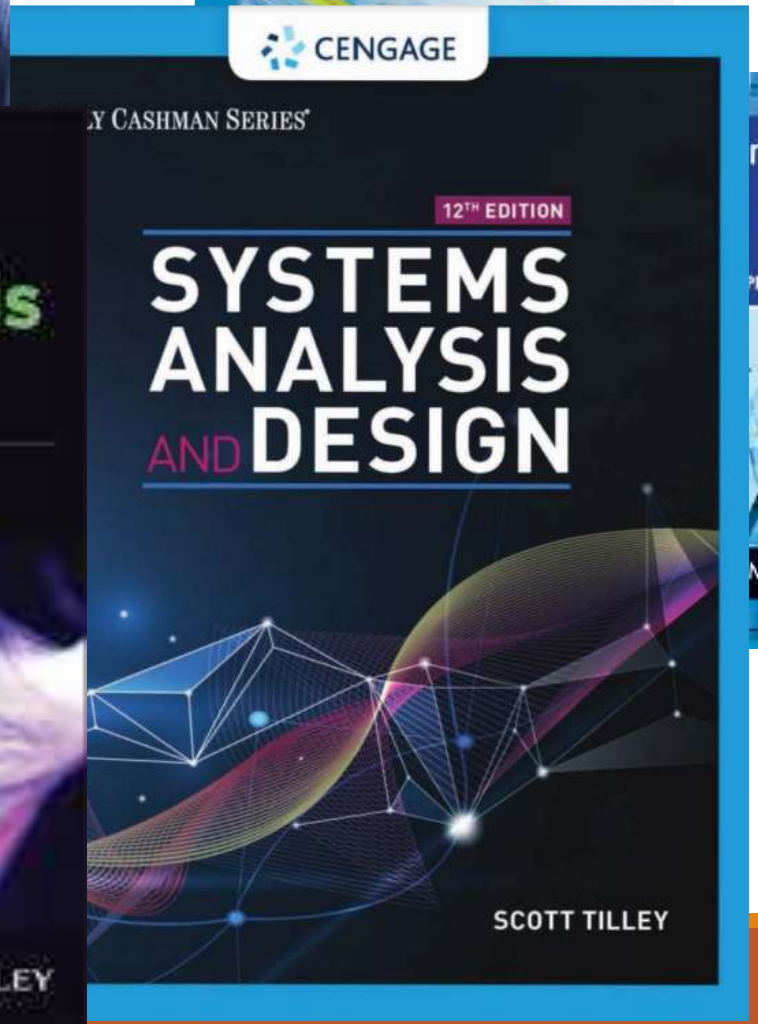

SYSTEMS ANALYSIS
& DESIGN


Systems Analysis
& Design

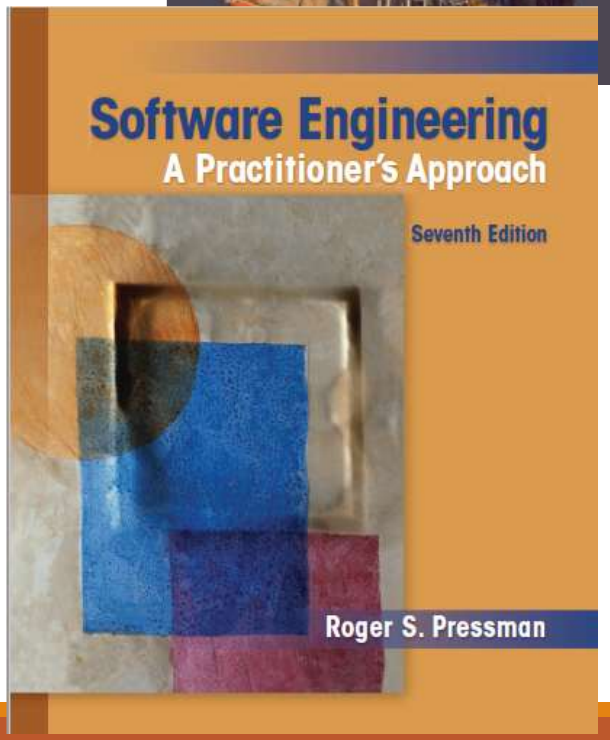

THIRD EDITION
Object-Oriented
Systems Analysis
And Design
Using UML


SOMMERVILLE
SOFTWARE ENGINEERING


Eighth Edition
Systems Analysis
and Design
Alan Dennis · Barbara Haley Wixom
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A Practitioner's Approach
Seventh Edition
Roger S. Pressman

What is a System?

The word System is derived from Greek word Systema, which means an organized relationship between any set of components to achieve some common cause or objective.

A system is "an orderly grouping of interdependent components linked together according to a plan to achieve a specific goal."

Constraints of a System

A system must have three basic constraints −

A system must have some **structure and behavior** which is designed to achieve a predefined objective.

Interconnectivity and **interdependence** must exist among the system components.

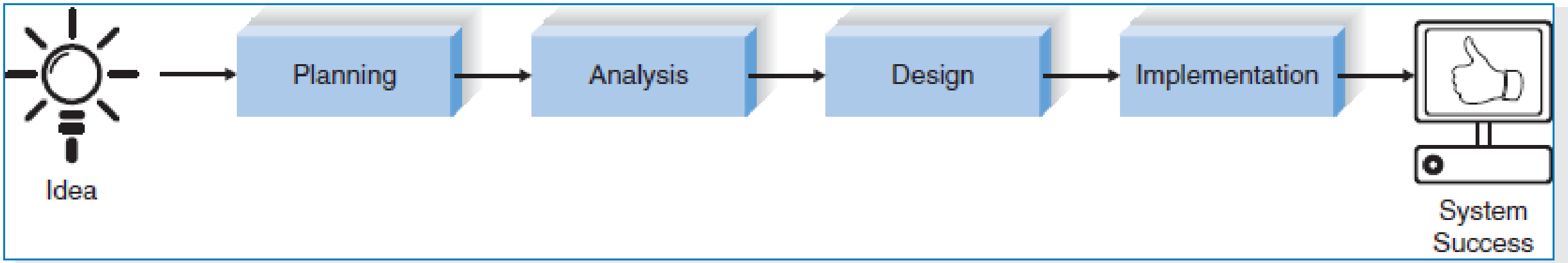
The **objectives of the organization** have a **higher priority** than the objectives of its subsystems.

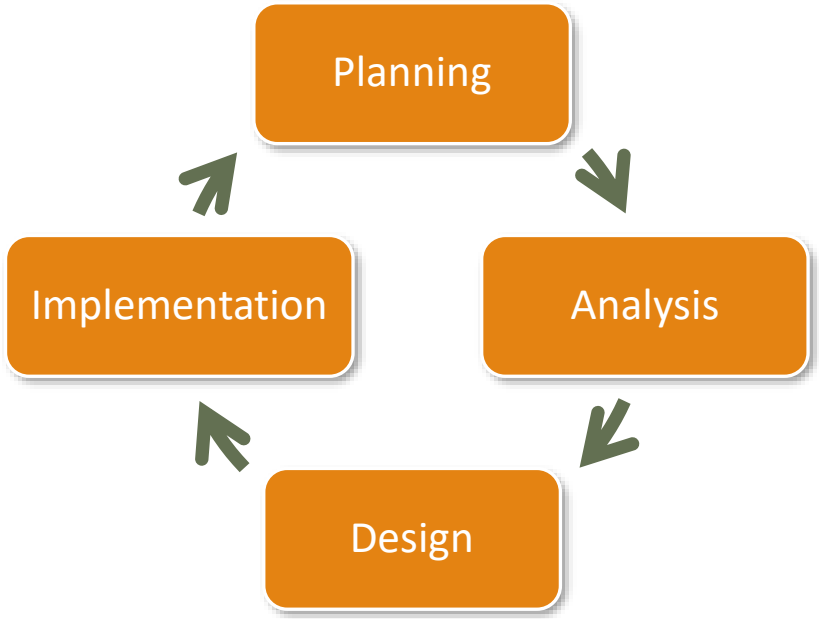
THE SYSTEMS DEVELOPMENT LIFE CYCLE

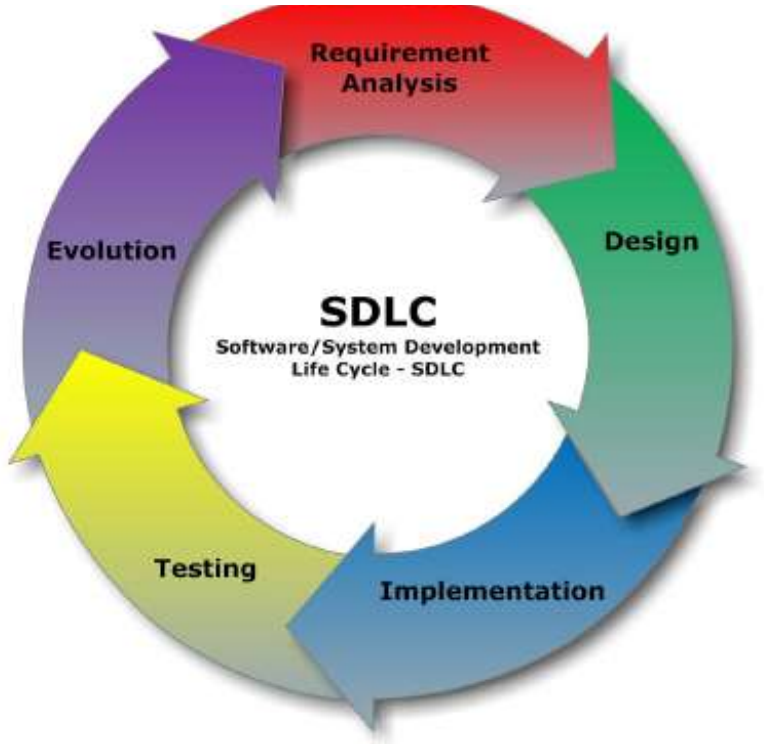
The **Systems Development Life Cycle (SDLC)** is the process of understanding how an information system (IS) can support business needs by designing a system, building it, and delivering it to users.

-”System Analysis Design UML Version 2.0”

THE SYSTEMS DEVELOPMENT LIFE CYCLE







SDLC: The Planning Phase

1. Project Initiation

- Develop/receive a system request
- Conduct a feasibility analysis
  - Technical feasibility
  - Economic feasibility
  - Organizational feasibility

2. Project Management

- Develop the work plan
- Staff the project
- Monitor & control the project

Phase	Chapter	Step	Technique	Deliverable
Planning	1	Identify opportunity	Project identification	System request
Focus: Why build this system?	1	Analyze feasibility	Technical feasibility Economic feasibility Organizational feasibility	Feasibility study
How to structure the project?	2	Develop workplan	Time estimation Task identification Work breakdown structure PERT chart Gantt chart Scope management	Project plan — work plan
Primary outputs: — System Request with feasibility study — Project plan	2	Staff project	Project staffing Project charter	— Staffing plan
	2	Control and direct project	CASE repository Standards Documentation Timeboxing Risk management	— Standards list — Risk assessment

SDLC: The Analysis Phase

1. Develop an analysis strategy
   - Model the current system
   - Formulate the new system
2. Gather the requirements
   - Develop a system concept
   - Create a business model to represent:
     - Business data
     - Business processes
3. Develop a system proposal

Phase	Chapter	Step	Technique	Deliverable
Analysis Focus: Who, what, where, and when for this system? Primary output — System proposal	3	Develop analysis strategy	Business process automation Business process improvement Business process reengineering	System proposal
	3	Determine business requirements	Interview JAD session Questionnaire Document analysis Observation	— Requirements definition
	4	Create use cases	Use case analysis	— Use cases
	5	Model processes	Data flow diagramming	— Process models
	6	Model data	Entity relationship modeling Normalization	— Data model

SDLC: The Design Phase

1. Develop a design strategy
2. Design architecture and interfaces
3. Develop databases and file specifications
4. Develop the program design to specify:
   - What programs to write
   - What each program will do

Phase	Chapter	Step	Technique	Deliverable
Design Focus: How will this system work? Primary output: — System specification	7	Design physical system	Design strategy	Alternative matrix System specification
	8	Design architecture	Architecture design Hardware & software selection	— Architecture report — Hardware & software specification
	9	Design interface	Use scenario Interface structure Interface standards Interface prototype Interface evaluation	— Interface design
	10	Design programs	Data flow diagramming Program structure chart Program specification	— Physical process model — Program design
	11	Design databases and files	Data format selection Entity relationship modeling Denormalization Performance tuning Size estimation	— Database & file specification — Physical data model

SDLC: The Implementation Phase

1. Construct the system
   - Build it (write the programming code)
   - Test it
2. Install system
   - Train the users
3. Support the system (maintenance)

Phase	Chapter	Step	Technique	Deliverable
Implementation Focus: delivery and support of completed system Primary output: — Installed system	12	Construct system	Programming Software testing Performance testing	Test plan Programs Documentation
	13	Install system	Conversion strategy selection Training	Migration plan — Conversion plan — Business contingency plan — Training plan
	13	Maintain system	Support selection System maintenance Project assessment	Support plan Problem report Change request
	13	Post-implementation	Post-implementation audit	Post-implementation audit report